

Week 5 – Workshop

Isostasy: Ocean–Continent Structure & Isostatic Gravity



Geophysics 3A: Potential Fields & Geothermics

July 6, 2026

Duration: 3 hours

Setting: Workshop room (small-group whiteboard work)

Preparation

Pre-reading:

Lecture videos:

Notes:

Purpose

This workshop develops a quantitative understanding of isostatic equilibrium and its gravity expression by combining column-based reasoning, ocean–continent structure, and slab gravity approximations. Airy and Pratt isostasy are treated as end-member models, with emphasis on why real Earth structure requires a mixed or layered approach involving the lithospheric mantle.

Learning Outcomes

By the end of this workshop, you should be able to:

- Explain isostatic equilibrium as equal lithostatic pressure (or mass per unit area) at a compensation depth.
- Derive and interpret Airy and Pratt isostasy as conceptual end members.
- Explain why crust-only isostatic models fail for ocean–continent systems.
- Derive the Airy isostatic gravity correction using slab approximations.
- Explain why gravity anomalies over high topography may be small.

Session Flow (180 min)

0–10 min: Warm-Up: Defining Isostasy

Individually, write a one-sentence definition of isostatic equilibrium. These definitions are compared in pairs and then within your group, with the goal of converging on a precise statement.

Target idea:

Isostatic equilibrium requires equal lithostatic pressure (or mass per unit area) at a compensation depth beneath adjacent columns.

Groups assign roles (facilitator, skeptic, recorder, assumption checker).

10–30 min: Mini-Lecture: Airy, Pratt, and Mixed Isostasy

A short instructor-led lecture reviews:

- Airy isostasy: variable crustal thickness, uniform density
- Pratt isostasy: constant thickness, lateral density variations
- Why neither end member fully describes the Earth
- The role of lithospheric mantle and partial compensation

This provides the conceptual framework for the two analytical activities that follow.

30–70 min: Activity A: Oceanic vs Continental Columns

Purpose

To test whether crust-only Airy isostasy can account for observed ocean–continent elevation differences, and to explore how including the lithospheric mantle resolves inconsistencies.

Crust-Only Airy Isostasy

Each group balances an oceanic and a continental column under the assumption that compensation occurs solely through crustal thickness. You will:

- Enforce equal mass per unit area above a compensation depth.
- Solve for the oceanic crust density required for equilibrium.
- Compare your result with realistic oceanic crust densities.

You should find that the required density is unrealistic, indicating a missing component in the model.

Two-Layer Lithosphere Extension

The model is then extended to include a lithospheric mantle beneath both oceanic and continental crust. You will:

- Rebalance the columns including mantle lithosphere thickness differences.
- Estimate the density contrast between oceanic and continental lithospheric mantle.
- Relate this contrast to mineralogical differences (e.g. garnet content).

70–80 min: Whole-Class Synthesis

A brief instructor-led discussion synthesises results from Activity 1, focusing on:

- Why crust-only Airy isostasy fails
- Why the lithospheric mantle must be included
- Where Pratt-like behaviour may arise

80–120 min: Activity B: Isostatic Gravity Correction

Purpose

To derive the Airy isostatic gravity correction and explain why large topography does not necessarily produce large gravity anomalies.

Airy Mass Balance

You will show that, under Airy isostasy,

$$\rho_c h = (\rho_m - \rho_c) r,$$

and derive an expression for the crustal root thickness $r(x)$.

Infinite Slab Gravity

Using the slab approximation,

$$\Delta g = 2\pi G \rho t,$$

you will compute the gravitational effects of surface topography and of the compensating crustal root.

Cancellation

By substitution, you will demonstrate that under perfect Airy compensation, the gravity effects of topography and root cancel exactly, and discuss what assumptions are required for this result.

120–145 min: Numerical Demonstration

The instructor presents a live Python demonstration illustrating:

- Airy compensation with varying compensation fractions
- Pratt-style lateral density variations
- Resulting gravity anomalies and residuals

Interpretation of results is emphasised rather than coding details.

145–170 min: Group Synthesis

Each group prepares a brief summary addressing:

- One reason crust-only Airy isostasy fails.
- One way lithospheric mantle structure resolves this failure.
- One equation linking topography, density, and gravity.
- One observation that could reduce non-uniqueness in isostatic interpretation.

170–180 min: Exit Questions

Individually, respond to the following:

1. Identify one unrealistic assumption in either Airy or Pratt isostasy and explain why it fails.
2. Explain how geological or geophysical constraints (e.g. seismology or petrology) help resolve ambiguity in isostatic models.

Activity A

Isostasy Problem: Oceanic vs Continental Crust

Part A: Crust-only Airy Isostasy

Given:

oceanic crust thickness: $h_{oc} = 7 \text{ km}$

ocean depth: $h_w = 3.7 \text{ km}$

seawater density: $\rho_w = 1030 \text{ kg m}^{-3}$

continental crust thickness: $h_{cc} = 40 \text{ km}$

continental crust density: $\rho_{cc} = 2850 \text{ kg m}^{-3}$

continental elevation: $\varepsilon = 0.8 \text{ km}$

mantle density: $\rho_a = 3330 \text{ kg m}^{-3}$

Instructions: Assume isostatic compensation occurs solely through crustal thickness. Compute the density of the oceanic crust, ρ_{oc} , that balances the oceanic and continental columns. Compare your result with typical oceanic crust densities ($\sim 2900\text{--}3000 \text{ kg m}^{-3}$) and discuss why the crust-only assumption may fail.

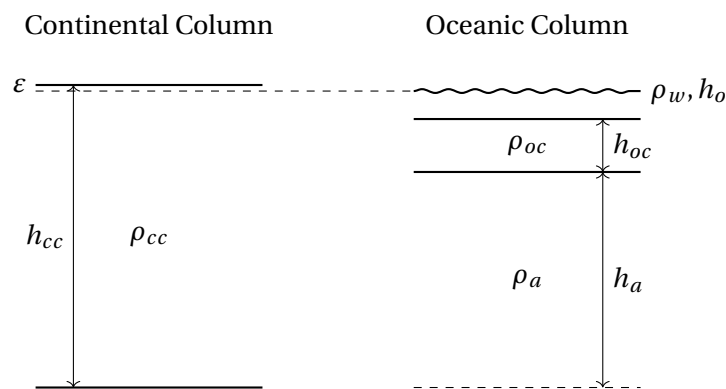


Figure 1: Comparison of isostatic columns for the continental and oceanic crust for Part A.

Questions:

1. Compute the required oceanic crust density ρ_{oc} .
2. Compare with expected oceanic crust densities and discuss why the simple crust-only model may not yield realistic values.

Part B: Two-layer Lithosphere Model

Given additional information:

Oceanic lithospheric mantle thickness $h_{om} = 75 \text{ km}$

Oceanic lithospheric mantle density: $\rho_{om} = 3330 \text{ kg m}^{-3}$

Continental lithospheric mantle thickness: $h_{cm} = 150 \text{ km}$

Continental lithospheric mantle density: $\rho_{cm} = ?$

Asthenosphere density: $\rho_a = 3320 \text{ kg m}^{-3}$

Instructions: Let's add a two-layered lithosphere using average lithospheric thicknesses for the oceanic and continental settings. Our goal will be to see what we can learn about the relative density difference between continental and oceanic mantle lithosphere. In the process, we'll show how including the mantle lithosphere resolves the unrealistic crust-only result.

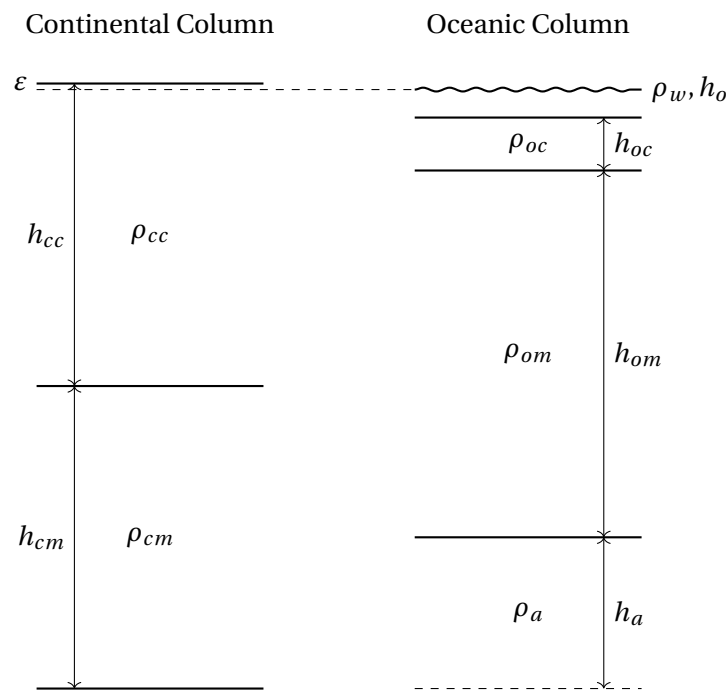


Figure 2: Comparison of isostatic columns for the continental and oceanic crust for Part B.

Table 1: Modal proportions of minerals in the oceanic and continental lithospheres.

Mineral	Oceanic	Continental
olivine	75	60
orthopyroxene	25	17
clinopyroxene	10	11
garnet	0	12

Oceanic lithospheric mantle is assumed to be composed of depleted peridotite with a density of $\rho_{om} = 3330 \text{ kg m}^{-3}$. Continental lithospheric mantle may contain a higher proportion of garnet.

Density of depleted mantle (peridotite): $\rho_d = 3330 \text{ kg m}^{-3}$

Density of garnet: $\rho_g = 3600 \text{ kg m}^{-3}$

Questions:

1. Setup the isostatic equations for a two-layered mantle lithosphere. If we assume an average oceanic crustal density $\rho_{oc} = 2900\text{--}3000 \text{ kg m}^{-3}$, what density of the continental mantle lithosphere?
2. Explain how including the mantle lithosphere resolves the unrealistic density obtained in the crust-only case.
3. Discuss the role of garnet fraction in the continental lithospheric mantle for the density contrast. Compute the percentage difference garnet between the oceanic and continental lithosphere and compare your result with the modal proportions in Table 1.

Activity B

Isostatic Gravity Correction

Purpose

In this problem, you will derive the isostatic gravity correction under the assumption of Airy isostasy, starting from a Bouguer slab formulation and progressing to a practical implementation using topography and a reference crustal thickness.

Background

Consider a reference Earth model with a constant crustal thickness T_0 . Surface topography $h(x)$ is compensated by a crustal root $r(x)$ such that columns of equal width have equal mass above a compensation depth.

Let:

- ρ_c = crustal density
- ρ_m = mantle density
- $h(x)$ = surface elevation
- $r(x)$ = additional root thickness relative to T_0

Geometry

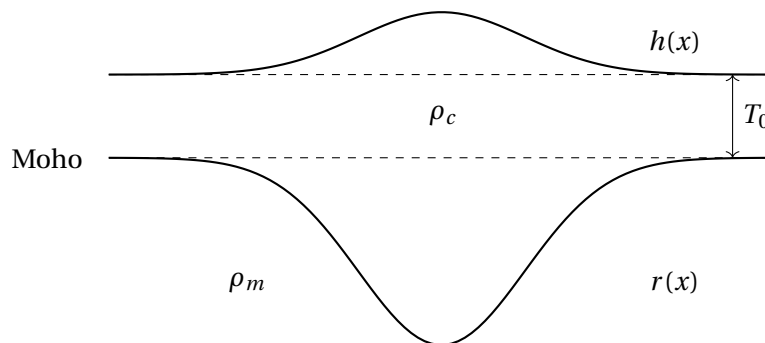


Figure 3: A mountain with a topographic expression $h(x)$ and compensating root $r(x)$.

Part A: Airy Isostasy

1. Starting from mass balance between a reference column and a column with topography, show that:

$$\rho_c h = (\rho_m - \rho_c) r$$

2. Solve for the root thickness $r(x)$ in terms of $h(x)$.

Part B: Gravity of an Infinite Slab

The gravitational attraction of an infinite horizontal slab of thickness t and density ρ is:

$$g = 2\pi G \rho t$$

3. Write an expression for the gravitational effect of the topography.
4. Write an expression for the gravitational effect of the crustal root. Carefully define the density contrast.

Part C: Substitution and Simplification

5. Substitute your expression for $r(x)$ into the root gravity term.
6. Show that the total gravity effect of the topography plus its Airy root simplifies to:

$$g_{\text{total}} = 0$$

Part D: Practical Isostatic Correction

In practice, crustal thickness is not known everywhere. Instead, we assume a reference thickness T_0 and compute $r(x)$ from observed topography.

7. Explain why the isostatic correction is model-dependent.
8. Write the final expression for the isostatic correction in terms of $h(x)$ only.

Target Result

You should show that the gravity effect of the compensating root is:

$$g_{\text{root}} = -2\pi G\rho_c h$$

and that, under perfect Airy compensation, the combined effect of topography and root cancels exactly.

Extension (Optional)

Discuss how this result would change if:

- compensation were incomplete
- the lithosphere behaved elastically (flexural isostasy)